

Léo Crampé

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Education

University of Southern California

Master of Science in Mechanical Engineering (Artificial Intelligence and Machine Learning Emphasis)

Bachelor of Science in Biomedical Engineering (Mechanical Emphasis)

GPA: 3.66/4.00

Relevant Coursework: Biomedical Computer Simulation Methods, Rehabilitation Engineering and Assistive Technologies, Electricity and Magnetism, Mechanics & Thermodynamics, Molecular Biology, Differential Equations

Los Angeles, CA

December 2027

May 2027

Anglo-American University

Study Abroad Exchange Program

Prague, Czechia

August 2023 - December 2023

Experience

Zimmer Biomet

Systems Design Intern

Montreal, QC

May 2025 - August 2025

- Tested the ROSA Knee System as part of the performance team focused on biomechanics analysis for cut accuracy evaluations, bone registration, knee state angles, and bone painting in accordance with ISO 13485 QMS and design validation requirements
- Reduced processing times from 3 minutes to 3 seconds across 200+ executions by developing a SolidWorks macro that automated alignment of femur and tibia models using 3D spatial coordinate algorithms for precise validation of robot calculated cut planes
- Authored sections of 7 FDA compliant DVP & DVFRs for medical device design, gaining experience in regulatory compliant documentation and traceability processes in a french speaking engineering environment

Department of Defense, NAVSEA

Student Engineer

Port Hueneme Naval Base, CA

August 2022 - December 2022

- Functioned as an informational analyst studying data from an underwater rover which interprets geophysical insights
- Prototyped initial designs for 20+ rockets with CAD, 3D printing, and military equipment in a defense engineering environment

Insead Business School for the World

Chief Operating Officer during Summer Program

Fontainebleau, France

July 2022 - August 2022

- Led as the Chief Operating Officer of a 6 member team in a simulated bicycle company, streamlining manufacturing by reducing design cost by 30% and implementing risk analysis and mitigation strategies to ensure product safety
- Established bold manufacturing and rebranding changes to differentiate ourselves from competitors and drive innovation

Projects

Smart Knee Brace

CAD Design Lead

Los Angeles, CA

January 2025 - February 2025

- Constructed an Arduino-powered smart knee brace utilizing Fusion360 to assist athletes in recovering from patellar tendonitis 2x faster by integrating vibration therapy and real-time planar knee motion-tracking using accelerometers
- Implemented haptic feedback alerts to notify users of risky movements, promoting safer rehabilitation

Computational Modeling of Action Potentials

Student Researcher

Los Angeles, CA

September 2025 - November 2025

- Characterized system behavior across 50+ simulations by developing an Excel and MATLAB based computational models of neuronal action potentials, analyzing voltage propagation, spike frequency, and conductance behavior under varying excitation and inhibition inputs to investigate system performance and variability

Electric Guitar

Electrical Design Lead

Los Angeles, CA

March 2025 - May 2025

- Developed and fabricated a custom electric guitar with an integrated active filter circuit to enhance tonal performance
- Designed and tested analog filters using op-amps and a charge pump for dual-voltage operation
- Created schematics and PCB layouts in KiCad, managing end-to-end circuit board production

Skills & Extracurriculars

Technical Skills: Solidworks, Fusion360, Siemens NX, AutoCAD, Autodesk Fusion, PTC Creo, CATIA V5, ICIDO, ENOVIA, Revit, TinkerCAD, MATLAB, Python, C/C++, Java, Arduino IDE, Labview, KiCad, LTspice, Microsoft Excel, Minitab, DFMEA, RCA, GMP, QA, ISO 13485 QMS, GD&T, FEA, GD&T, DFMA, 3D Printing, PCR, Digital signal processing, Soldering, Welding, Oscilloscope, Strain gauge, Load cell, Accelerometer, Thermocouple, Pitot tube, FTIR, Gyroscope

Languages: Fully Bilingual in English and French

Organizations: Theta Tau Professional Engineering Society, Engineers Without Borders, MEDesign, Ski Team, Brazilian Jiu Jitsu

International Exposure: US and French Citizen, raised in Singapore; studied/worked in the U.S., Canada, France, and Czechia